

# Ehsan Naghdbishi

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## Professional Summary

*Chemical Engineer with 5+ years of applied experience spanning process simulation, laboratory materials analysis, and environmental process engineering for water and wastewater treatment (WWTP). Engineered a novel bio-adsorbent through Design of Experiments (DOE) and Response Surface Methodology (RSM) that improved pollutant removal performance across variable concentration ranges, and built a zero-waste cellulose recycle-flow reactor system that converts spent coffee grounds into functional adsorbents for textile wastewater treatment. Author of four peer-reviewed and conference-presented studies – including a first-author publication in the International Journal of Biological Macromolecules – and improved academic performance from a 14.49 to a 16.6/20 GPA between degrees, reflecting steady growth in technical rigor and independent problem-solving. Proficient in Aspen HYSYS, Pipesim, MATLAB, and Design Expert, and ready to bring hands-on process design, biosorption, and materials expertise to an R&D or process engineering team in environmental biotechnology, sustainable materials, or industrial wastewater treatment.*

## Professional Experience

### Laboratory Expert

*Baspar Payaesh Pars Knowledge-Based Company – Mashhad, Iran*

**November 2021 – April 2022**

- **Material Testing & Validation:** Executed and validated analytical procedures across polymeric, resin, and bio-based material batches, delivering accurate physical and chemical property data that directly supported client-facing R&D deliverables.
- **Quality Assurance:** Evaluated key performance characteristics of raw and processed material batches, strengthening quality-control decision-making within a knowledge-based industrial R&D framework.
- **Process Documentation:** Authored structured laboratory reports and analytical records that standardized internal QA protocols, improving reproducibility and audit-readiness across polymer and composite material evaluation.

### Process Intern

*East Oil and Gas Exploitation Company – Khangiran, Iran*

**June 2020 – September 2020**

- **Facility Operations Training:** Completed structured on-site training across upstream and midstream gas processing facility operations, building a systems-level understanding of industrial process design and safety protocols.
- **Pipeline Flow Simulation:** Built and ran gas pipeline flow simulations in **Pipesim**, developing hands-on proficiency in simulation-based process modeling for hydrocarbon transport systems.

## Key Technical Projects

### Bio-Adsorbent Design for Water Pollutant Removal

*Master's Thesis, Ferdowsi University of Mashhad*

**July 2023 – February 2025**

*Design and Optimization of a Novel Bio-Adsorbent for Water Pollutant Removal via DOE/RSM*

- **Biosorbent Engineering:** Designed and iteratively optimized a novel, biologically derived adsorbent material to solve persistent water pollutant removal challenges, applying biosorption isotherm and kinetic theory to guide material development from concept to validated performance.
- **Statistical Process Optimization:** Applied **Design Expert** and Response Surface Methodology (RSM) to statistically model and optimize adsorption parameters, validating robust performance across variable pollutant concentration ranges – a methodology directly transferable to industrial process scale-up.
- **Cost-Effective Materials Substitution:** Delivered a low-cost, bio-based adsorbent alternative to conventional activated carbon, reducing material input costs while meeting environmental treatment performance targets.

## Zero-Waste Biorefinery Valorization of Spent Coffee Grounds

*Applied Research – Cellulose Production & Textile Wastewater Treatment*

**2024 – 2025**

- **Process Engineering:** Engineered a cellulose production process from spent coffee grounds using a series recycle-flow reactor configuration, maximizing raw material yield while minimizing process waste generation.
- **Waste-Stream Upcycling:** Converted reactor effluent into functional bio-based adsorbent materials for textile wastewater treatment, closing the process loop into a fully zero-waste valorization pathway and eliminating a secondary waste stream.
- **Feedstock Flexibility:** Investigated a supplementary plant-based cellulose extraction method from coffee waste, broadening feedstock flexibility for future process scale-up and reducing single-source supply risk.

## Industrial Microbiology & Bioreactor Process Design

- **Fermentation Process Design:** Applied industrial-level fermentation engineering and industrial microbiology principles to biologically driven wastewater treatment systems, improving process reliability for scale-up applications.
- **Nanoparticle-Enhanced Bioprocessing:** Investigated silica and graphene oxide nanoparticle integration to enhance sulfide pollutant removal efficiency within activated sludge bioreactors, strengthening treatment outcomes for industrial wastewater.
- **Kinetic Modeling & Publication:** Built and validated kinetic models in **MATLAB** characterizing pollutant degradation rates, culminating in a peer-reviewed journal publication and demonstrating quantitative modeling rigor applicable to process control and optimization work.

## Drilling Fluid Reuse & Wastewater Residue Treatment

- **Acidic Treatment Study:** Evaluated the effect of acidic treatment and reuse of spent drilling fluid on the biological treatability of resulting wastewater residues, presenting findings at a national engineering conference.

## Materials & Composite Systems

- **Polymer & Resin Chemistry:** Applied hands-on knowledge of polymers, paints, and advanced resin compounds gained through direct laboratory testing and characterization work.
- **Composite Part Fabrication:** Delivered industrial-level design and implementation of composite parts, integrating materials science principles with practical manufacturing constraints.

## Education

### M.Sc. in Chemical Engineering (Biotechnology)

*Ferdowsi University of Mashhad – Mashhad, Iran*

**Oct 2022 – Feb 2025**

*GPA: 16.6 / 20*

### B.Sc. in Chemical Engineering

*Shahrood University of Technology – Shahrood, Iran*

**Oct 2016 – Sep 2021**

*GPA: 14.49 / 20*

## Publications & Conference Papers

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### A Zero-Waste Approach to Spent Coffee Grounds Valorization: Cellulose Production in Series Recycle-Flow Reactors with Its Effluent Upcycling into Adsorbents for Textile Wastewater Treatment

*International Journal of Biological Macromolecules*

**2025**

*DOI: 10.1016/j.ijbiomac.2025.147444 – Presents a closed-loop, zero-waste process converting spent coffee grounds into cellulose via series recycle-flow reactors, with reactor effluent upcycled into functional adsorbents for textile wastewater treatment.*

### Kinetic Modeling of Sulfide Pollutant Removal in Activated Sludge Bioreactor with Presence of Silica and Graphene Oxide Nanoparticles

*Journal of Water and Wastewater (wwcerd.com)*

**Ref: WWJ-2108-3173**

*Developed a biosorption kinetic model characterizing sulfide pollutant degradation within an activated sludge bioreactor, quantifying the catalytic and adsorptive contribution of silica and graphene oxide nanoparticles.*

### Effect of Acidic Treatment and Reuse of Spent Drilling Fluid on Wastewater Treatment Residues

*4th National Conference on Numerical & Experimental Methods in Chemical and Mining Industries (CEC1403)*  
**Feb 2025**

*Conference Link – Investigated acid-based treatment and reuse strategies for spent drilling fluid to reduce the environmental burden of resulting wastewater treatment residues.*

### Extraction of Plant Cellulose from Coffee Waste

*18th National Congress of Chemical Engineering Iran*

**Oct 2024**

*Certificate Link – Presented an alternative plant-based cellulose extraction methodology derived from coffee waste feedstock.*

## Technical & Professional Competencies

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| <b>Process Design:</b>            | Process Analysis & Design (Advanced), Water & Wastewater Treatment (WWTP) Process Design (Advanced), Pollutant Adsorbent Design & Implementation (Advanced), Bioreactor Engineering & Scale-Up                               |
| <b>Biotechnology:</b>             | Industrial Microbiology, Fermentation Process Design (Industrial), Biosorption Kinetics & Isotherm Modeling, Environmental Biotechnology, Applied Biotechnology                                                              |
| <b>Materials Science:</b>         | Polymers, Paints & Advanced Resins, Composite Part Design & Implementation (Industrial), Sustainable & Bio-Based Materials                                                                                                   |
| <b>Research Methods:</b>          | Design of Experiments & Response Surface Methodology (DOE/RSM), Kinetic & Statistical Modeling, Peer-Reviewed Technical Writing                                                                                              |
| <b>Simulation &amp; Software:</b> | Aspen HYSYS (Intermediate), Pipesim (Intermediate), MATLAB (Advanced), Design Expert – DOE/RSM (Advanced), Python (Intermediate), Microsoft Office (Advanced), Adobe Photoshop (Intermediate), Adobe Premiere (Intermediate) |
| <b>Professional Skills:</b>       | Analytical Precision, Laboratory Rigor, Cross-Disciplinary Problem Solving, Independent Research Initiative, Continuous Improvement                                                                                          |

## Languages

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**Persian** (Native)    **English** (Advanced)

## Interests

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Water & Wastewater Treatment | Bio-Adsorbent Materials | Industrial Microbiology | Zero-Waste Valorization  
| Nanomaterial-Enhanced Bioprocesses | Environmental Biotechnology